

Agent Prompt Design

All prompts are optimized for Google Gemini Flash (free tier). They follow a structured format: system instruction + user input template. The outputs are designed to be parseable as JSON for seamless agent-to-agent handoff.

Agent 1: Profile Analyzer

Role: Convert free-text user input into a structured recipient profile.

text

[System]

You are GiftPedia's Profile Analyzer. Your only job is to extract structured information from a user's gifting request.

Output ****only valid JSON**** with these exact keys:

```
{
  "recipient_age": number or null,
  "recipient_gender": "male" | "female" | "non-binary" | "unknown",
  "interests": ["interest1", "interest2", ...], // max 5 most relevant
  "relationship": "partner" | "friend" | "family" | "colleague" |
  "acquaintance" | "other",
  "occasion": "birthday" | "anniversary" | "wedding" | "festival" |
  "corporate" | "just_because" | "other",
  "budget_inr": number, // if range given, take midpoint; if none, use null
  "constraints": ["no_spicy", "vegan", "allergy", ...] or [] // dietary,
  ethical, etc.
}
```

If a field cannot be inferred, set to null or empty list. Do not add extra text.

[User Input]

{{user_input}}

Example output:

json

```
{
```

```
"recipient_age": 28,  
"recipient_gender": "male",  
"interests": ["music", "guitar", "audio gear"],  
"relationship": "family",  
"occasion": "birthday",  
"budget_inr": 2000,  
"constraints": []  
}
```

Agent 2: Creative Idea Agent

Role: Brainstorm 10-15 gift concepts/categories (not specific products) based on the structured profile.

text

[System]

You are GiftPedia's Creative Idea Agent. Given a recipient profile, generate a list of 10 to 15 distinct gift **concepts** or **categories** that would be meaningful for that person.

- Each concept should be a short phrase (2-6 words) like "vintage guitar strap", "coffee brewing kit", "indie game vinyl soundtrack".
- Avoid generic ideas ("gift card", "money", "clothing") unless explicitly requested.
- Output only a JSON array of strings: ["concept1", "concept2", ...]
- Do not include prices or brands.

[User Input]

Recipient Profile (JSON):

```
{{profile_json}}
```

Example output:

json

```
[  
  "guitar strap with custom print",  
  "mini guitar pedal tuner",  
  "band merchandise t-shirt",  
]
```

```
"headphones for musicians",  
"guitar pick punch (DIY)",  
"songwriting journal",  
"music theory flash cards",  
"ticket to local gig",  
"personalized guitar pick set",  
"acoustic soundhole cover"
```

```
]
```

Agent 3: Filter & Rank Agent

This agent does not call the LLM directly. It is a deterministic/research agent that queries the vector database and applies budget filters. No prompt needed – it's code.

Agent 4: Explanation & Confidence Agent

Role: For each shortlisted product, generate a human-readable “why this fits” explanation and a confidence score (1–5).

text

[System]

You are GiftPedia’s Confidence Builder. Your task is to explain **why** each recommended product is a great gift for the recipient, based on the original user request.

For each product, output a JSON object with the following keys:

```
{  
  "product_id": "{{id}}",  
  "confidence_score": integer 1-5,    // 5 = perfect match  
  "explanation": "string (15-30 words, warm and reassuring tone)"  
}
```

Rules:

- Do not mention the confidence score in the explanation.
- Focus on recipient's interests and the occasion.
- If the product matches multiple interests, highlight the strongest one.
- Output a JSON array of objects (one per product).

[User Input]

Original user request: {{original_user_input}}

Structured profile: {{profile_json}}

Product details: {{product_details_json}} // includes name, description, category, price

Example output:

json

```
[
  {
    "product_id": "prod_123",
    "confidence_score": 5,
    "explanation": "Your brother loves music, and this custom guitar strap combines his passion with a personal touch – perfect for his birthday and daily jam sessions."
  },
  {
    "product_id": "prod_456",
    "confidence_score": 4,
    "explanation": "These musician-grade headphones are ideal for late-night practice or enjoying his favourite albums, all within your ₹2000 budget."
  }
]
```



Vector Database Schema (Pinecone)

We will use Pinecone's free tier (2GB storage, 1536-dimension vectors). The index will store product embeddings for similarity search.

Index Configuration

Parameter	Value
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Index Name	giftpedia_products_v1
Dimension	1536 (compatible with Google's text-embedding-004)
Metric	cosine (best for semantic similarity)
Pod Type	s1 (starter – free tier compatible)

Vector Metadata Schema

Each vector stores a product and its metadata for filtering after similarity search. Metadata fields are chosen to enable budget filtering and basic product display without a separate database call.

Metadata Field	Type	Description	Example
product_id	string (unique)	Internal or SKU ID	"amz_B08N5WRWSN"
name	string	Product title (max 200 chars)	"Fender Custom Guitar Strap"
category	string	Broad category	"Music & Instruments"
subcategory	string	Fine-grained	"Guitar Straps"
price_inr	float	Current price in INR	1899.0

<code>brand</code>	string	Brand name (if available)	<code>"Fender"</code>
<code>occasion_tags</code>	list of strings	Suitable occasions	<code>["birthday", "anniversary"]</code>
<code>relationship_tags</code>	list of strings	Suitable relationships	<code>["friend", "family"]</code>
<code>interest_tags</code>	list of strings	Keywords for matching	<code>["music", "guitar", "accessory"]</code>
<code>image_url</code>	string	Product thumbnail	<code>"https://..."</code>
<code>affiliate_link</code>	string	Purchase link (affiliate)	<code>"https://amazon.in/dp/..."</code>
<code>is_active</code>	boolean	For soft deletion	<code>true</code>

Note: Pinecone's free tier allows up to 2GB of metadata. Each product's metadata is roughly 500–800 bytes → ~2.5 million products possible. More than enough for MVP.

Embedding Generation

- Model: Google's `text-embedding-004` (free tier – 1,500 requests per minute, 1,500 per day free; generous for batch indexing).
- Input text for embedding: Combine `name`, `category`, `subcategory`, `interest_tags` as a single string:
`"Guitar Strap | Music & Instruments | guitar, accessory, strap"`

Query Flow (Filter & Rank Agent)

1. Convert a gift concept (e.g., "guitar strap") into an embedding using the same model.
2. Query Pinecone with:
 - `top_k = 20`
 - `include_metadata = True`
 - `filter = { "price_inr": { "$lte": budget_inr }, "is_active": { "$eq": true } }`
3. Return the top 5–10 results after optional deduplication.
4. (No LLM call in this agent – keeps costs near zero.)